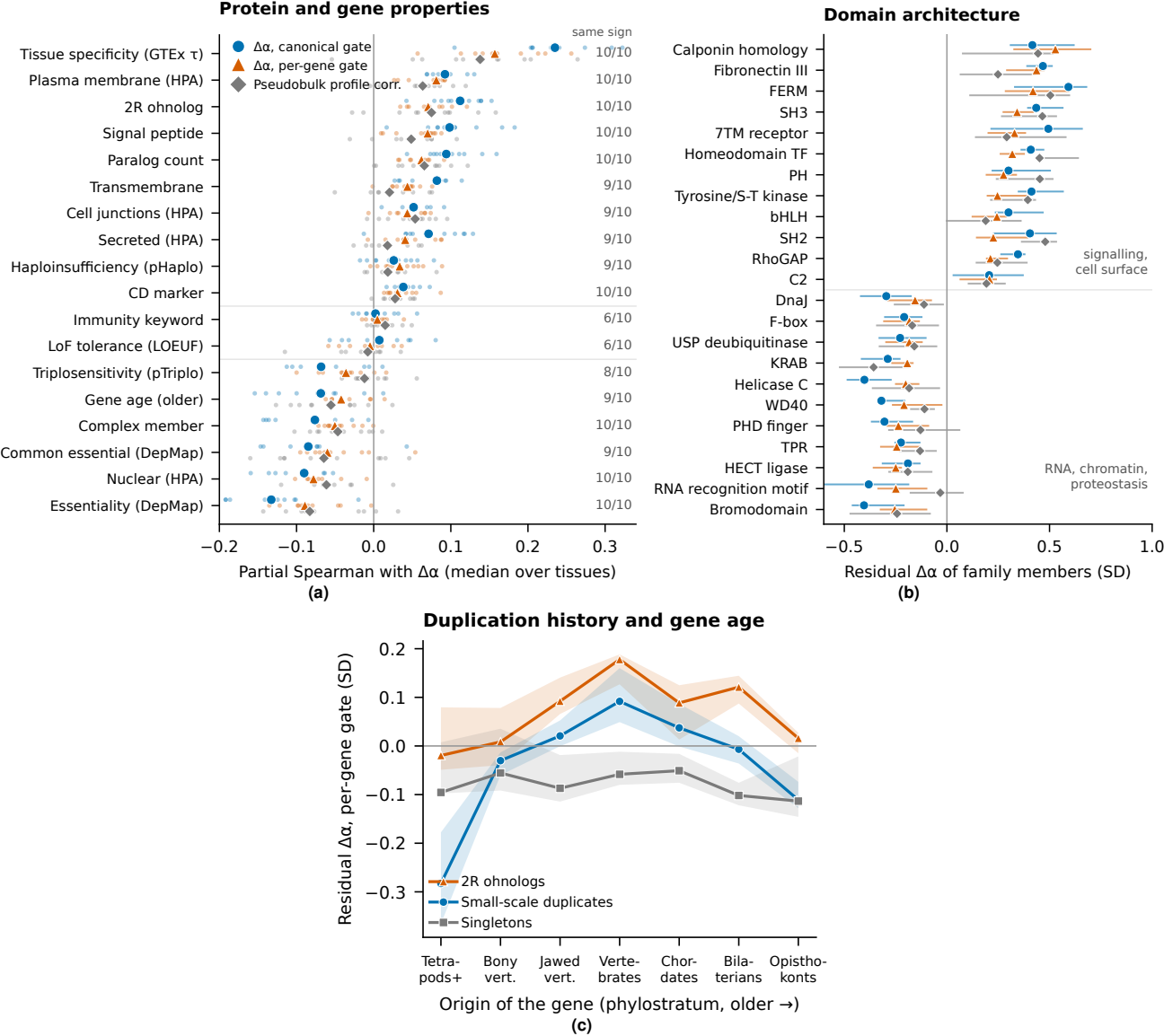

Protein and gene properties
same sign
Δα, canonical gate
Δα, per-gene gate
Pseudobulk profile corr.
Tissue specificity (GTEx τ)
Plasma membrane (HPA)
2R ohnolog
Signal peptide
Paralog count
Transmembrane
Cell junctions (HPA)
Secreted (HPA)
Haploinsufficiency (pHaplo)
CD marker
Immunity keyword
LoF tolerance (LOEUF)
Triplosensitivity (pTriplo)
Gene age (older)
Complex member
Common essential (DepMap)
Nuclear (HPA)
Essentiality (DepMap)
10/10
10/10
10/10
10/10
10/10
9/10
9/10
9/10
9/10
10/10
6/10
6/10
8/10
9/10
10/10
9/10
10/10
10/10
−0.2
−0.1
0.0
0.1
0.2
0.3
Partial Spearman with Δα (median over tissues)
(a)
Domain architecture
Calponin homology
Fibronectin III
FERM
SH3
7TM receptor
Homeodomain TF
PH
Tyrosine/S-T kinase
bHLH
SH2
RhoGAP
C2
DnaJ
F-box
USP deubiquitinase
KRAB
Helicase C
WD40
PHD finger
TPR
HECT ligase
RNA recognition motif
Bromodomain
signalling, cell surface
RNA, chromatin, proteostasis
−0.5
0.0
0.5
1.0
Residual Δα of family members (SD)
(b)
Duplication history and gene age
Residual Δα, per-gene gate (SD)
0.2
0.1
0.0
−0.1
−0.2
−0.3
2R ohnologs
Small-scale duplicates
Singletons
Tetra-pods+
Bony vert.
Jawed vert.
Verte-brates
Chor-dates
Bila-terians
Opistho-konts
Origin of the gene (phylostratum, older →)
(c)